

MATH3026/4022 Lecture 15

Linear Filtering

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- ▶ Linear filtering in time series: aims and objectives
- ▶ Application of linear filters to time series in the frequency domain
- ▶ Interpretation of the effect of a filter in the filtered time series
- ▶ Examples

Introduction

In Lecture 13, we considered how time series could be described and defined using the frequency domain.

A natural question would be 'why consider a time series in the frequency domain when it might seem more intuitive to use the time domain?'

In answer, if we wish to perform a linear transformation of a time series (as we often might) it is easier to do this by first considering the series in frequency domain.

As we shall see, it is straightforward to recover the ACF, PACF etc. of the transformed series when we use the frequency domain.

Linear Filtering – Time Series

Why would we want to apply a linear transformation to a time series?

In answer, most real-world, time series data are **noisy**.

By this, we mean that the observed time series data will contain some elements that we want to know about (for example, a linear trend over time) but the data will also contain ‘noise’ – information about which we don’t wish to make inference.

A linear filter may be applied to a time series to amplify (and recover) important features of the series (e.g. general trends) and reduce the influence of unwanted components (e.g. random error).

Linear Filter

A **linear filter** is simply a **linear transformation** applied to a time series.

For example, consider the stationary time series $\{Y_t: t \in \mathbb{Z}\}$.

Suppose that a linear transformation of the following form is applied to Y_t

$$X_t = \sum_{j=-\infty}^{\infty} b_j Y_{t-j} \quad \text{with} \quad \sum_{j=-\infty}^{\infty} b_j^2 < \infty.$$

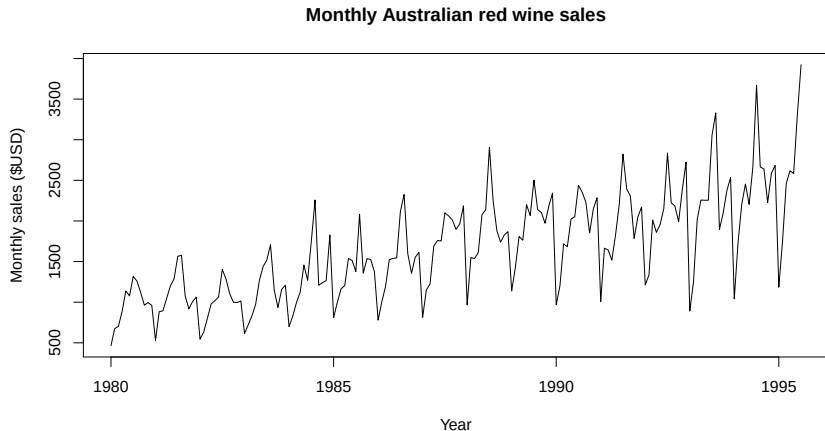
For example

$$X_t = \frac{1}{3} Y_{t-1} + Y_t + \frac{2}{3} Y_{t+1}$$

We say that X_t is a linear filtered version of Y_t .

Example 15.1: Australian Red Wine Sales

From Lecture 1, recall the time series of monthly Australian red wine sales from 1980 to 1994.



Example 15.1: Australian Red Wine Sales

We observe that there appears to be a general linear trend (with wine sales increasing with time).

However, the time plot is not 'smooth' because of the obvious seasonality in the data.

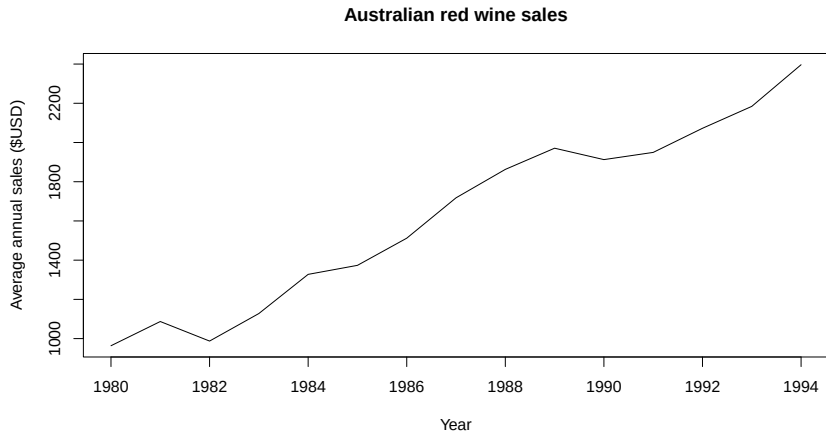
We might wish to **smooth** the data to recover a linear trend by taking the annual average over the 12 months of a calendar year.

That is, define the time series $\{X_t\}$ to be such that

$$X_t = \text{Sample mean of sales data from year } t$$

Example 15.1: Australian Red Wine Sales

A time plot of the annual Australian red wine sales time series is shown below.



Example 15.1: Australian Red Wine Sales

Clearly, the annual average time series is smoother than the monthly average time series.

What we've done here, is apply a **linear transformation** to our original time series that has **smoothed** the original series.

We can now observe the linear trend – if that is what is of interest – more easily than before.

However, we note that – in this example – we've reduced the number of data points substantially and we would not necessarily want to do this.

A moving average (covered later) might be better for smoothing.

Linear Filter: Frequency Domain

As we shall see, it is easiest to consider linear filtering using time series written in the frequency domain.

We begin by deriving some important results.

From Lecture 13, we know that, using the frequency domain, Y_t can be written

$$Y_t = \int_{-\pi}^{\pi} e^{iwt} dS_y(w);$$

and

$$X_t = \int_{-\pi}^{\pi} e^{iwt} dS_x(w).$$

Linear Filter: Frequency Domain

Linear Filter: Frequency Domain

Linear Filter: Frequency Domain

We deduce that

$$g_X(w) = |h(w)|^2 g_Y(w)$$

In words, we see that we can use the spectral density function of Y_t to obtain the spectral density function of X_t .

This spectral density function can then tell us all that we need to know about the correlation structure etc. of the linear filtered series, X_t . Here

- ▶ $h(w)$ is called the **transfer function**.
- ▶ $|h(w)|$ is called the **gain**.
- ▶ $|h(w)|^2$ is called the **squared gain**.

Examples

The transfer function has the form

$$h(w) = \sum_{j=-\infty}^{\infty} b_j e^{-i w j}.$$

For the remainder of the lecture, we shall explore various examples of linear filters and how these filters transform features of the original time series.

Throughout, $\{Y_t\}$ will denote the **original time series** and $\{X_t\}$ will denote the **transformed time series**.

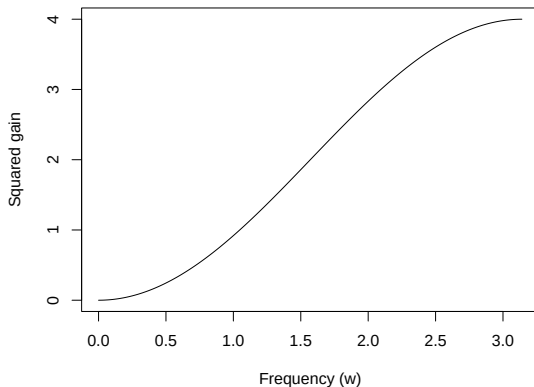
We'll begin by considering one of the most common and basic filters – the first difference of a time series, written

$$X_t = Y_t - Y_{t-1}.$$

Example 15.2: First Difference

Example 15.2: First Difference

A plot of the frequency against the squared gain is shown below.



Example 15.2: First Difference

Since $g_X(w) = |h(w)|^2 g_Y(w)$, we see that

$g_X(w)$ is small where w is small.

$g_X(w)$ is large as $w \rightarrow \pi$.

In words, this implies that the first difference amplifies high frequency but dampens down low frequency.

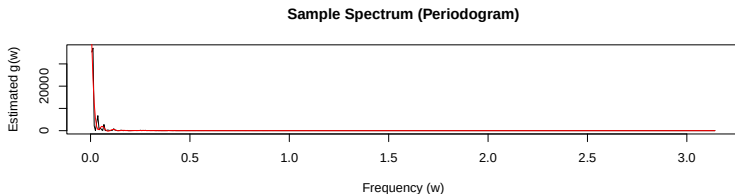
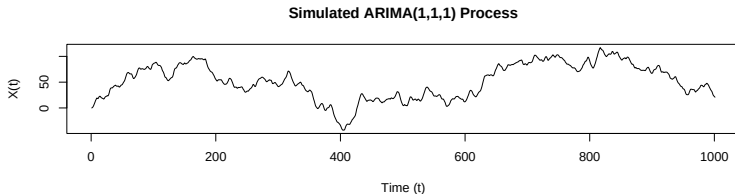
This means that trends (low frequency) are removed but the transformed series will be less smooth (as higher frequencies are amplified).

In general, we should avoid over-differencing as we see that this would induce higher frequencies into the transformed time series.

Example 15.2: First Difference

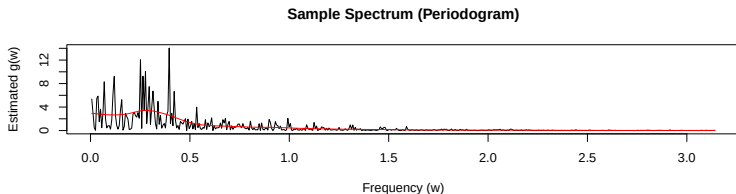
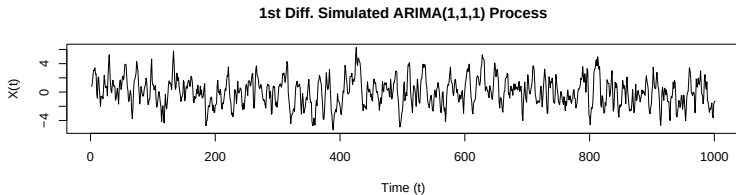
To see this behaviour, we simulate 1000 observations from an $ARIMA(1, 1, 1)$ process with $\phi = 0.7$ and $\theta = 0.5$.

A time plot and periodogram for these data are shown below.



Example 15.2: First Difference

Now, we produce similar plots for the first differences of these data.



Example 15.3: Lag 12 Differencing

We have seen that lag 12 differencing is often used where data are collected monthly and there is seasonality over a yearly cycle.

For lag 12 differencing,

$$\begin{aligned} X_t &= \sum_j b_j Y_{t-j} \text{ with } b_0 = 1, b_{12} = -1 \text{ and } b_j = 0 \text{ otherwise.} \\ &= Y_t - Y_{t-12}. \end{aligned}$$

The transfer function is

$$\begin{aligned} h(w) &= \sum_j b_j e^{-i w j} \\ &= 1 - e^{-12 i w}. \end{aligned}$$

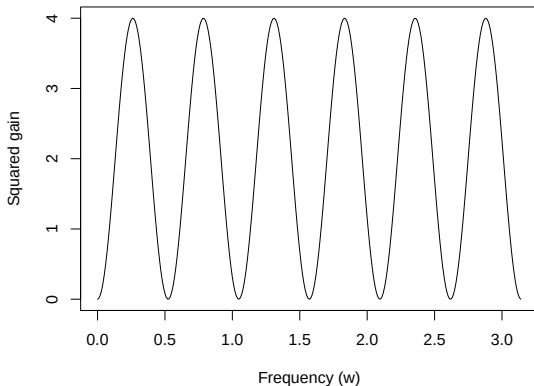
Example 15.3: Lag 12 Differencing

The squared gain is

$$\begin{aligned}|h(w)|^2 &= |1 - e^{-12iw}|^2 \\&= (1 - e^{-12iw})(1 - e^{-12iw})^* \\&= (1 - e^{-12iw})(1 - e^{12iw}) \\&= 2 - (e^{12iw} + e^{-12iw}) \\&= 2(1 - \cos 12w) \\&= 4 \sin^2 6w \quad (\text{because } \cos 2\theta = 1 - 2 \sin^2 \theta).\end{aligned}$$

Example 15.3: Lag 12 Difference

A plot of the frequency against the squared gain for the lag 12 difference is shown below.



Example 15.3: Lag 12 Difference

From the plot of the squared gain, we see that differencing at lag 12 will remove frequencies

$$\frac{\pi k}{6} \text{ for } k = 0, 1, \dots, 6.$$

At these frequencies, the period is given by

$$\text{Period} = \frac{2\pi}{w} = \frac{12\pi}{\pi k} = \frac{12}{k}.$$

We conclude that differencing at lag 12 will remove oscillations of period 12, 6, 4, 3, 12/5 and 2.

Example 15.4: Moving Average Filter

The moving average filter is defined

$$X_t = \frac{1}{2m+1} \sum_{j=-m}^m Y_{t-j}.$$

In words, this filter takes an average of the values of time series, in the time range

$$\{t-m, \dots, t-1, t, t+1, \dots, t+m\}.$$

and is sometimes known as the mean filter. Formally

$$X_t = \sum_j b_j Y_{t-j}$$

with

$$b_j = \begin{cases} \frac{1}{2m+1} & j \in \{-m, \dots, m\}; \\ 0 & \text{otherwise.} \end{cases}$$

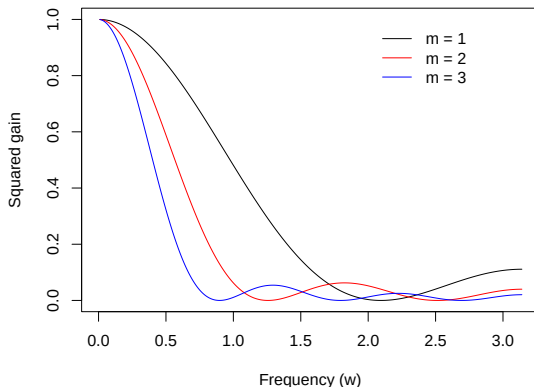
Example 15.4: Moving Average Filter

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Example 15.4: Moving Average Filter

The squared gain is

$$|h(w)|^2 = \frac{1}{(2m+1)^2} \frac{\sin^2\left(\frac{2m+1}{2}w\right)}{\sin^2\left(\frac{w}{2}\right)}$$



Example 15.4: Moving Average Filter

Here we see that high frequency is removed and low frequency amplified.

In addition, frequencies $\frac{2\pi k}{2m+1}$ for $k = 1, 2, \dots$ are removed completely.

Moving average filters may be useful for the removal of high frequency noise from a time series.

Example 15.5: Repeated Average Filter

Suppose that $Y_t^{(0)}$ is a white noise process. We define

$$\begin{aligned}Y_t^{(1)} &= \frac{1}{2} \left(Y_t^{(0)} + Y_{t-1}^{(0)} \right) \\Y_t^{(2)} &= \frac{1}{2} \left(Y_t^{(1)} + Y_{t-1}^{(1)} \right) \\&\vdots \\Y_t^{(k)} &= \frac{1}{2} \left(Y_t^{(k-1)} + Y_{t-1}^{(k-1)} \right)\end{aligned}$$

So, to transform the series $Y_t^{(k)}$

$$Y_t^{(k+1)} = \sum_j b_j Y_{t-j}^{(k)} \quad \text{with } b_0 = 1/2, b_1 = 1/2 \text{ and } b_j = 0 \text{ otherwise.}$$

We define $g_k(w) = g_{Y^{(k)}}(w)$ as the spectral density of $Y_t^{(k)}$.

Example 15.5: Repeated Average Filter

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We see that

$$\begin{aligned} g_2(w) &= |h(w)|^2 g_1(w) \\ &= |h(w)|^4 g_0(w). \end{aligned}$$

In general

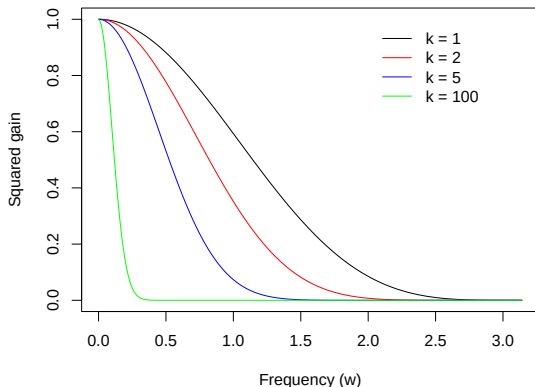
$$\begin{aligned} g_k(w) &= (|h(w)|^2)^k g_0(w) \\ &= \left(\frac{1 + \cos w}{2} \right)^k \frac{\sigma_z^2}{2\pi}. \end{aligned}$$

We see that applying the filter k times is equivalent to filtering the original series $Y_t^{(0)}$ with transfer function:

$$\left(\frac{1 + \cos w}{2} \right)^{\frac{k}{2}}.$$

Example 15.5: Repeated Average Filter

A plot of the squared gain against the frequency for the repeated average filter, for different values of k , is given below



Example 15.5: Repeated Average Filter

We see that low frequencies are amplified more for larger values of k .

If k is very large (i.e. $k \rightarrow \infty$) then

$$\left(\frac{1 + \cos w}{2} \right)^k \rightarrow 0$$

except where $w = 0$. Hence, the spectral density for very large k would be such that

$$g_k(w) \approx \begin{cases} \frac{\sigma_z^2}{2\pi} & w = 0; \\ 0 & \text{otherwise.} \end{cases}$$

In other words, we would smooth the series so much that we'd end up with $X_t^{(k)} = c$ for some constant c .

Example 15.6: Repeated Difference Filter

Suppose that $Y_t^{(0)}$ is a white noise process. We define

$$\begin{aligned}Y_t^{(1)} &= \frac{1}{2} \left(Y_t^{(0)} - Y_{t-1}^{(0)} \right) \\Y_t^{(2)} &= \frac{1}{2} \left(Y_t^{(1)} - Y_{t-1}^{(1)} \right) \\&\vdots \\Y_t^{(k)} &= \frac{1}{2} \left(Y_t^{(k-1)} - Y_{t-1}^{(k-1)} \right)\end{aligned}$$

So, to transform the series $Y_t^{(k)}$

$$X_t = \sum_j b_j Y_{t-j}^{(k)} \quad \text{with } b_0 = 1/2, b_1 = -1/2 \text{ and } b_j = 0 \text{ otherwise.}$$

We define $g_k(w) = g_{Y^{(k)}}(w)$ as the spectral density of $Y_t^{(k)}$.

Example 15.6: Repeated Difference Filter

The transfer function is

$$h(w) = \frac{1}{2}(1 - e^{-iw}).$$

The squared gain is given by

$$\begin{aligned}|h(w)|^2 &= \frac{1}{4}(1 - e^{-iw})(1 - e^{iw}) \\ &= \frac{1}{4}(2 - 2\cos w) \\ &= \frac{1}{2}(1 - \cos w).\end{aligned}$$

Since $\{Y_t^{(0)}\}$ is white noise, $g_0(w) = \frac{\sigma_z^2}{2\pi}$. It follows that

$$g_1(w) = |h(w)|^2 g_0(w) = \frac{\sigma_z^2}{4\pi}(1 - \cos w).$$

Example 15.6: Repeated Difference Filter

We see that

$$\begin{aligned} g_2(w) &= |h(w)|^2 g_1(w) \\ &= |h(w)|^4 g_0(w). \end{aligned}$$

In general

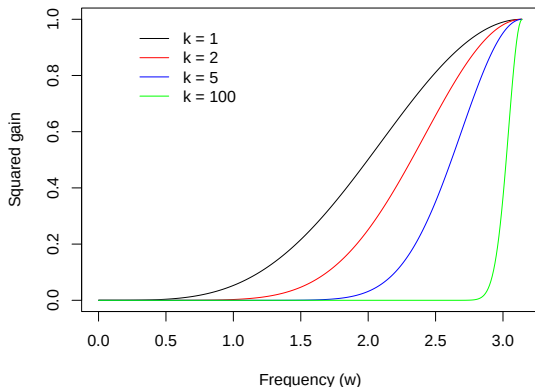
$$\begin{aligned} g_k(w) &= (|h(w)|^2)^k g_0(w) \\ &= \left(\frac{1 - \cos w}{2} \right)^k \frac{\sigma_z^2}{2\pi}. \end{aligned}$$

We see that applying the filter k times is equivalent to filtering the original series $\{Y_t^{(0)}\}$ with transfer function:

$$\left(\frac{1 - \cos w}{2} \right)^{\frac{k}{2}}.$$

Example 15.6: Repeated Difference Filter

A plot of the squared gain against the frequency for the repeated difference filter, for different values of k , is given below



Example 15.6: Repeated Difference Filter

We see that higher frequencies are amplified increasingly as k increases.

If k is very large then

$$g_k(w) = \begin{cases} \frac{\sigma_z^2}{2\pi} & w = \pi; \\ 0 & \text{otherwise.} \end{cases}$$

The period is then equal to 2.

Hence, if k is very large we'd obtain a time series with terms of equal magnitude, but where successive terms would have different signs.

Example 15.7: General Filter

Suppose that the time series $\{Y_t\}$ is an $ARMA(p, q)$ process and we recall that

$$\phi(B)Y_t = \theta(B)Z_t$$

where $\{Z_t\}$ is a white noise process.

Now suppose that $X_t = C(B)Y_t$ is a filtered version of Y_t using some function $C(B)$. Then

$$\begin{aligned}X_t &= C(B)Y_t \\&= C(B)\frac{\theta(B)}{\phi(B)}Z_t \\&= \psi(B)Z_t\end{aligned}$$

where $\psi(B) = C(B)\frac{\theta(B)}{\phi(B)}$.

Example 15.7: General Filter

The autocovariance generating function is given by

$$\begin{aligned}\Gamma_X(z) &= \sigma_z^2 \psi(z) \psi(z^{-1}) \\ &= \sigma_z^2 C(z) \frac{\theta(z)}{\phi(z)} C(z^{-1}) \frac{\theta(z^{-1})}{\phi(z^{-1})} \\ &= C(z) C(z^{-1}) \Gamma_Y(z)\end{aligned}$$

where $\Gamma_Y(z)$ is the acgf for the time series Y_t .

We know that

$$\begin{aligned}g_X(w) &= \frac{1}{2\pi} \Gamma_X(e^{-iw}) \\ &= \frac{1}{2\pi} C(e^{-iw}) C(e^{iw}) \Gamma_Y(e^{-iw}) \\ &= C(e^{-iw}) C(e^{iw}) g_Y(w) \\ &= |h(w)|^2 g_Y(w) \quad \text{where } |h(w)|^2 = C(e^{-iw}) C(e^{iw}).\end{aligned}$$

Example 15.7: General Filter

For example, suppose

$$\begin{aligned} X_t &= \frac{1}{6} [-Y_{t-2} + 2Y_{t-1} + 4Y_t + 2Y_{t+1} - Y_{t+2}] \\ \implies C(B) &= \frac{1}{6} [-B^2 + 2B + 4 + 2B^{-1} - B^{-2}] \end{aligned}$$

The squared gain is given by

$$\begin{aligned} |h(w)|^2 &= C(e^{-iw})C(e^{iw}) \\ &= \frac{1}{6} [-e^{-i2w} + 2e^{-iw} + 4 + 2e^{iw} - e^{i2w}] \\ &\quad \times \frac{1}{6} [-e^{i2w} + 2e^{iw} + 4 + 2e^{-iw} - e^{-i2w}] \\ &= \frac{1}{36} (4 - 2\cos 2w + 4\cos w)^2 \\ &= \frac{1}{9} (2 - \cos 2w + 2\cos w)^2 \end{aligned}$$

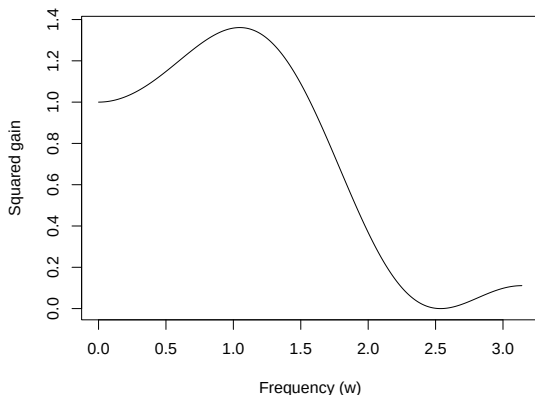
Example 15.7: General Filter

$$\begin{aligned}|h(w)|^2 &= \frac{1}{9}(2 - \cos 2w + 2 \cos w)^2 \\&= \frac{1}{9} [2 - (2 \cos^2 w - 1) + 2 \cos w]^2 \\&= \frac{1}{9} \left[-2 \left(\cos^2 w - \cos w - \frac{3}{2} \right) \right]^2 \\&= \frac{4}{9} \left[\left(\cos w - \frac{1}{2} \right)^2 - \frac{7}{4} \right]^2 \\&= \frac{4}{9} \left[\frac{7}{4} - \left(\cos w - \frac{1}{2} \right)^2 \right]^2\end{aligned}$$

We see that the squared gain, $|h(w)|^2$, is maximised where $\cos w = 1/2$. That is, where $w = \pi/3$.

Example 15.7: General Filter

This filter will enhance frequencies close to $\pi/3$ and reduce high frequencies.



Lecture 15: Learning Outcomes

- ▶ Understand the definition and concept of a **linear filter** and know why such filters are applied to time series data (e.g. for smoothing).
- ▶ Know how to apply linear filters to time series in the frequency domain.
- ▶ Know the definitions of the
 - ▶ Transfer function;
 - ▶ Gain and squared gain.
- ▶ Understand how to interpret the effect of a linear filter/transform by producing a plot of the frequency against the squared gain.